

Instructions:

Please provide your ID number, do not sign your name!

This is a four hour exam. Your solutions should be legible and clearly organized, written in complete sentences in good mathematical style on your own paper. All work should be your own—no outside sources are permitted—using methods and results from the first year topology courses. Each problem is worth the same number of points.

- (1) Consider the 2-sphere S^2 and the 2-dimensional torus T .
 - (a) For what values $m \in \mathbb{Z}$ does there exist a degree m map $f: S^2 \rightarrow T$?
 - (b) For what values $n \in \mathbb{Z}$ does there exist a degree n map $f: T \rightarrow S^2$?
 In both questions, if you state that a map of a certain degree exists then give a construction; otherwise prove its non-existence.

- (2) Let M be a smooth manifold and let $f: M \rightarrow \mathbb{R}$ be a smooth function. Define a linear map $D_f^k: \Omega^k(M) \rightarrow \Omega^{k+1}(M)$ by:

$$D_f^k(\omega) = d\omega + df \wedge \omega$$

where d is the standard exterior derivative.

- (a) Show that $D_f^{k+1} \circ D_f^k = 0$.
 - (b) Let ω be a k -form. Show that $D_f^k(\omega) = 0$ if and only if the form $e^f \omega$ is closed: $d(e^f \omega) = 0$.
 - (c) Prove that the map $\omega \mapsto e^f \omega$ defines a chain isomorphism between the chain complexes $(\Omega^*(X), D_f)$ and $(\Omega^*(X), d)$.
- (3) Let $F: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the map defined by $F(x, y, z) = (xz, yz, z)$. Let $f: S^2 \rightarrow \mathbb{R}^3$ be the restriction of F to the unit sphere S^2 . Determine the set of points $p \in S^2$ at which f fails to be an immersion.

- (4) Let G be a Lie group. Show that its tangent bundle TG is trivial, that is isomorphic to $G \times \mathbb{R}^n \rightarrow G$ for some n . (Recall that a Lie group G is a smooth manifold which is also a group, such that the multiplication map $(f, g) \mapsto f \cdot g$, and the inverse map $f \mapsto f^{-1}$ are smooth.)

Hint. Given $g \in G$, the map $L_g: G \rightarrow G$, given by $L_g(h) = gh$, may be helpful; note that it is a diffeomorphism.

- (5) Let X be a 2-dimensional disk with n distinct points on its boundary identified, $n \geq 2$. Let A be its boundary $\bigvee_{i=1}^n S^1$.
 - (a) Compute $\pi_1(X)$ and $\pi_1(A)$.
 - (b) Show that there is no retraction $r: X \rightarrow A$.
- (6) Let $X = (S^1 \vee S^1) \times (S^1 \vee S^1)$.

- (a) Compute $\pi_1(X)$.
 - (b) Compute the homology groups of X .
 - (c) Show that every continuous map $\mathbb{R}P^2 \rightarrow X$ is nullhomotopic.
- (7) For all $n \geq 0$, compute the homology groups with $\mathbb{Z}/2$ -coefficients of the space X obtained from the standard n -simplex Δ^n by identifying all faces of the same dimension.